

PRACTICAL-4

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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

CODE:

```
import pandas as pd
```

```
# Take inputs from the user to create a list of numbers numbers =  
list(map(int, input().split()))
```

```
# Create a Pandas series from the list of numbers series =  
pd.Series(numbers)
```

```
# Grouping by even and odd numbers and calculating the mean grouped  
=series.groupby(series % 2 == 0).mean()
```

```
# Display the mean of even and odd numbers with labels
grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

print("Mean of even and odd numbers:") print(grouped)
```

4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

CODE:

```
import pandas as pd
```

```
# Provided dictionary of lists data =
```

```
{  
    'Name': ['Alice', 'Bob', 'Charlie'],  
    'Age': [25, 30, 35],  
}
```

```
# Convert the dictionary to a DataFrame df =  
pd.DataFrame(data)
```

```
# Display the original DataFrame  
print("Original DataFrame:") print(df)
```

```
# Adding a new row new_name =  
input("New name: ") new_age =  
int(input("New age: ")) df.loc[len(df)] =  
[new_name, new_age]
```

```
# Display the DataFrame after adding a new row print("After  
adding a row:\n",df)
```

```
# Modifying a row row_to_modify = int(input("Index of  
row to modify: ")) new_age_value = int(input("New  
age: ")) df.at[row_to_modify, "Age"] = new_age_value
```

```
# Display the DataFrame after modifying a row  
print("After modifying a row:") print(df)
```

```
# Deleting a row row_to_delete = int(input("Index of row to  
delete: ")) df =  
df.drop(index=row_to_delete).reset_index(drop=True)
```

```
# Display the DataFrame after deleting a row print("After  
deleting a row:")
```

```
print(df)
```

```
# Adding a new column genders = input("Enter genders  
separated by space: ").split() df["Gender"]=genders
```

```
# Display the DataFrame after adding a new column
```

```
print("After adding a new column:") print(df)
```

```
# Modifying a column df["Name"] =
```

```
df["Name"].str.upper()
```

```
# Display the DataFrame after modifying a column
```

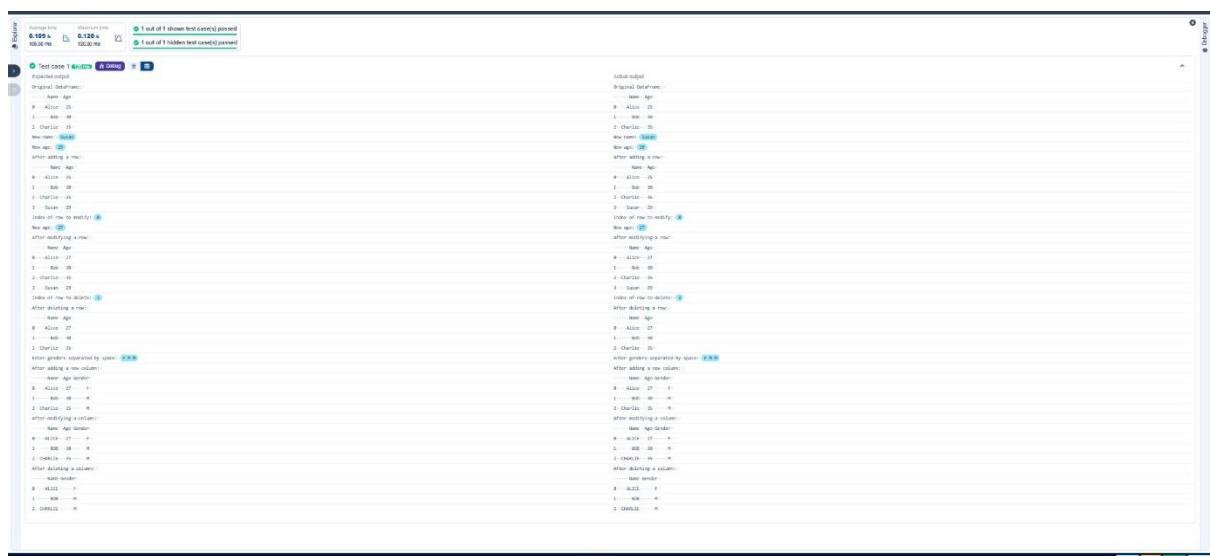
```
print("After modifying a column:") print(df)
```

```
# Deleting a column df =
```

```
df.drop(columns=["Age"])
```

```
# Display the DataFrame after deleting a column
```

```
print("After deleting a column:") print(df)
```



4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

CODE:

```
import pandas as pd
```

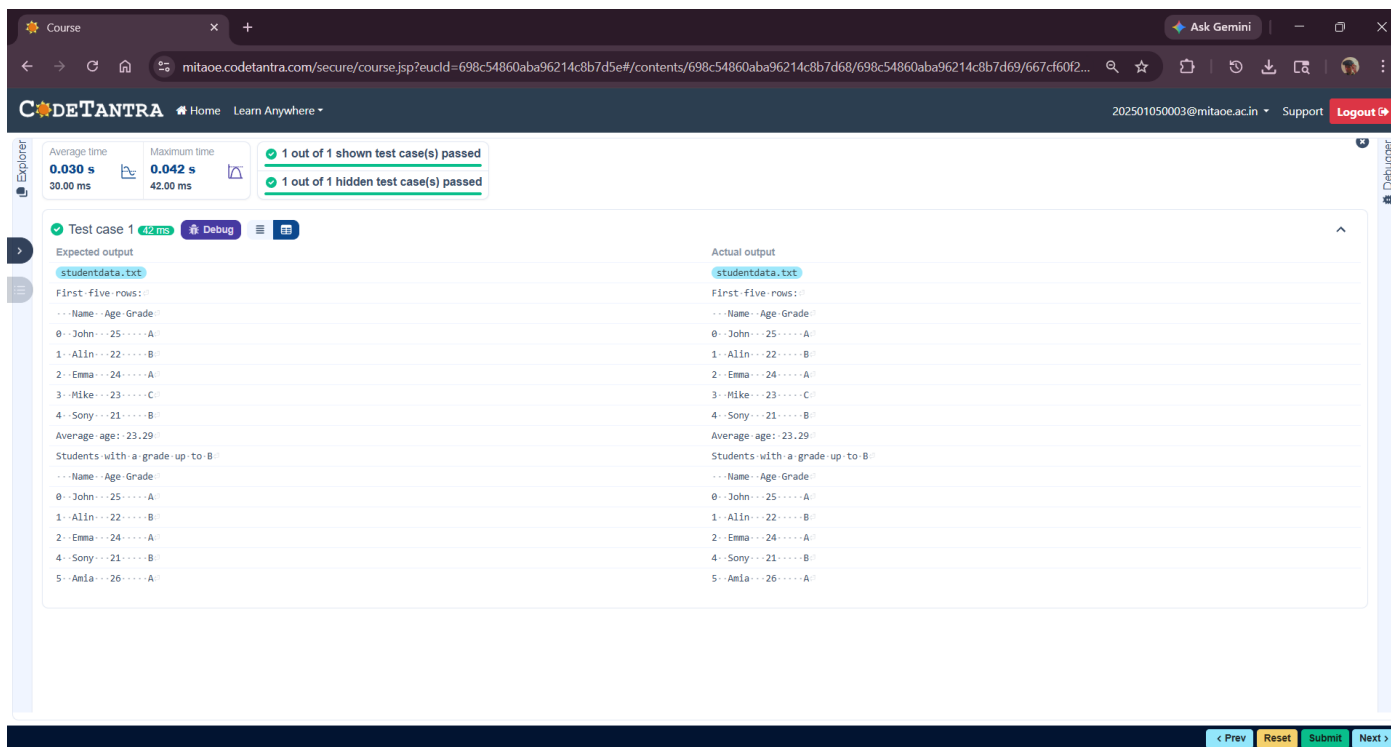
```
# Read the text file into a DataFrame file = input() data = pd.read_csv(file, sep="\s+",  
header=None, names=["Name", "Age", "Grade"])
```

```
# write your code here..
```

```
print("First five rows:") print(data.head())
```

```
average_age = round(data["Age"].mean(), 2) print(f"Average  
age: {average_age}")
```

```
filtered_students = data[data["Grade"]<= "B"] print("Students  
with a grade up to B") print(filtered_students)
```



4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago

2025-01-05,Product C,10,30,Los Angeles

CODE:

```
import pandas as pd
```

```
# Prompt the user for the file name file_name =  
input()
```

```
# Load the data df =  
pd.read_csv(file_name) df["Date"] =  
pd.to_datetime(df["Date"])
```

```
df["Month"] = df["Date"].dt.to_period("M")
```

```
df["Total Sales"] = df["Quantity"] * df["Price"]
```

```
monthly_sales = df.groupby("Month")["Total Sales"].sum() # Find the month with the highest total sales  
best_month = monthly_sales.idxmax() highest_sales = monthly_sales.max()
```

```
print(f"Best month: {best_month}") print(f"Total  
sales: ${highest_sales:.2f}")
```

Course

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Average time: 0.020 s (20.00 ms) Maximum time: 0.036 s (36.00 ms)

1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 36 ms Debug

Expected output	Actual output
sales_data.csv	sales_data.csv
Best month: 2025-01	Best month: 2025-01
Total sales: \$1210.00	Total sales: \$1210.00

< Prev Reset Submit Next >

4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago 2025-01-03,Product

B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago CODE:

```
import pandas as pd
```

```
# Prompt the user for the file name file_name =
```

```
input()
```

```
# Load the data df =
```

```
pd.read_csv(file_name)
```

```
product_sales = df.groupby("Product")["Quantity"].sum() #
```

```
Find the product with the highest total quantity sold
```

```
best_product = product_sales.idxmax() highest_quantity =
```

```
product_sales.max()
```

```
# Display the result print(f"Best selling product:
```

```
{best_product}") print(f"Total quantity sold:
```

```
{highest_quantity}")
```

The screenshot displays the CodeTANTRA web application interface. At the top, the browser address bar shows the URL: `mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d68/698c54860aba96214c8b7d6a/67861370...`. The CodeTANTRA logo and navigation links (Home, Learn Anywhere) are visible in the header. The user is logged in as `202501050003@mitaoe.ac.in`. The main content area shows the results of a test case execution. A summary bar at the top indicates: "Average time: 0.014 s (14.00 ms)", "Maximum time: 0.028 s (28.00 ms)", and "1 out of 1 shown test case(s) passed" and "2 out of 2 hidden test case(s) passed". Below this, a detailed view for "Test case 1" (28 ms) is shown. It compares the "Expected output" and "Actual output" for the file `sales_data.csv`. Both outputs are identical, showing: "Best-selling product: Product-A" and "Total quantity sold: 23". The interface includes a "Debugger" sidebar on the right and navigation buttons ("Prev", "Reset", "Submit", "Next") at the bottom right.

4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago 2025-01-

05,Product C,10,30,Los Angeles CODE:

```
import pandas as pd
```

```
# Prompt the user for the file name file_name =  
input()
```

```
# Load the data df = pd.read_csv(file_name)  
city_sales = df.groupby("City")["Quantity"].sum()  
best_city = city_sales.idxmax() # write the code..
```

```
# Display the result print(f"City sold the most  
products: {best_city}")
```

The screenshot displays the CodeTANTRA web application interface. At the top, the browser address bar shows the URL: `mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d68/698c54860aba96214c8b7d6a/67861553...`. The CodeTANTRA logo and navigation links (Home, Learn Anywhere) are visible. The user's email `202501050003@mitaoe.ac.in` and a Logout button are in the top right.

The main content area shows test results for a program. A summary box indicates: "1 out of 1 shown test case(s) passed" and "2 out of 2 hidden test case(s) passed". Below this, "Test case 1" is expanded, showing a duration of 28 ms and a "Debug" button. The "Expected output" and "Actual output" sections both display: `sales_data.csv` and `City sold the most products: Los Angeles`.

At the bottom of the interface, there are navigation buttons: "< Prev", "Reset", "Submit", and "Next >".

4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the following columns: Date, Product, Quantity, Price, and City.
- For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).
- Output the product pair/s that was sold most frequently.

Sample Data:

Date,Product,Quantity,Price,City

2025-01-01,Product A,5,20,New York

2025-01-01,Product B,3,15,Los Angeles

2025-01-02,Product A,7,20,New York

2025-01-02,Product C,4,30,Chicago

2025-01-03,Product B,2,15,Chicago

2025-01-03,Product A,8,20,Los Angeles

2025-01-04,Product C,6,30,New York

2025-01-04,Product B,5,15,Los Angeles

2025-01-05,Product A,3,20,Chicago 2025-01-05,Product

C,10,30,Los Angeles **Explanation:**

Transactions:

- **2025-01-01:** Product A, Product B
- **2025-01-02:** Product A, Product C
- **2025-01-03:** Product B, Product A
- **2025-01-04:** Product C, Product B
- **2025-01-05:** Product A, Product C

Now, let's count how often the pairs of products appear together:

- **Product A and Product B:** Appear in transactions on 2025-01-01 and 2025-0103.
- **Product A and Product C:** Appear in transactions on 2025-01-02 and 2025-0105.
- **Product B and Product C:** Appears in transactions on 2025-01-04.

Most Frequent Product Combinations:

- **Product A and Product B** (2 times)
- **Product A and Product C** (2 times)

CODE:

```
import pandas as pd from itertools
```

```
import combinations from
```

```
collections import Counter
```

```
# Prompt user to input the file name file_name =  
input()
```

```
# Read data from the specified CSV file df =  
pd.read_csv(file_name)
```

```
# write the code grouped =
```

```
df.groupby("Date")["Product"].apply(list)
```

```
pairs_counter = Counter() for products in grouped:
```

```
    pairs = combinations(sorted(products),2)    pairs_counter.update(pairs)
```

```
max_frequency = max(pairs_counter.values()) most_common_pairs =[(pair,freq) for  
pair, freq in pairs_counter.items() if freq == max_frequency]
```

```
# Output the most frequent product pairs for
```

```
(product1,product2),frequency in most_common_pairs:
```

```
    print(f"{product1} and {product2}: {frequency} times")
```

The screenshot displays the CodeTANTRA web application interface. At the top, the browser address bar shows the URL: `mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d68/698c54860aba96214c8b7d6a/6788eea8...`. The CodeTANTRA logo and navigation links (Home, Learn Anywhere) are visible. The user is logged in as `202501050003@mitaoe.ac.in`. The interface shows test results for a file named `sales_data.csv`. The top status bar indicates: "Average time: 0.015 s (14.67 ms)", "Maximum time: 0.028 s (28.00 ms)", and "1 out of 1 shown test case(s) passed" and "2 out of 2 hidden test case(s) passed". Below this, a table compares the "Expected output" and "Actual output" for "Test case 1" (28 ms). Both outputs are identical, showing "Product A and Product B: 2 times" and "Product A and Product C: 2 times". The bottom navigation bar includes buttons for "< Prev", "Reset", "Submit", and "Next >".

4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows of the dataset.
3. Get the shape of the dataset (number of rows and columns).
4. Get a summary of the dataset (using `.info()`).
5. Get basic statistics (mean, standard deviation, etc.) of the dataset using `.describe()`.
6. Check for missing values and display the count of missing values for each column.
7. Fill missing values in the 'Age' column with the median age.
8. Fill missing values in the 'Embarked' column with the most frequent value (`mode()`).
9. Drop the 'Cabin' column due to many missing values.
10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

CODE:

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset data =
```

```
pd.read_csv('Titanic-Dataset.csv')
```

```
# 1. Display the first 5 rows of the dataset print(data.head())
```

2. Display the last 5 rows of the dataset `print(data.tail())`

3. Get the shape of the dataset `print(data.shape)`

4. Get a summary of the dataset (info) `print(data.info())`

5. Get basic statistics of the dataset `print(data.describe())`

6. Check for missing values `print(data.isnull().sum())`

7. Fill missing values in the 'Age' column with the median age `median_age = data['Age'].median()`
`data['Age'].fillna(median_age,inplace= True)`

8. Fill missing values in the 'Embarked' column with the mode `mode_embarked =data['Embarked'].mode() [0]`
`data['Embarked'].fillna(mode_embarked, inplace = True)`

9. Drop the 'Cabin' column due to many missing values `data.drop('Cabin', axis=1, inplace=True)`

10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'

The screenshot shows a Jupyter Notebook with the following content:

- Cell 1:** Imports pandas and loads the Titanic dataset.


```
import pandas as pd
train = pd.read_csv('train.csv')
```
- Cell 2:** Displays the first 10 rows of the dataset.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund	male	22	1	0	1	5.49	C	S
2	1	1	Mr. Allen	male	35	1	0	24	53.1	C	S
3	1	3	Mr. Borchers	male	39	1	0	33	51.66	C	S
4	1	1	Mr. Brown	male	54	1	0	1	5.00	C	S
5	0	3	Mr. Carter	male	20	1	0	1	5.28	C	S
6	1	1	Mr. Fisher	male	30	1	0	1	12.36	C	S
7	1	3	Mr. Gibson	male	25	1	0	1	7.25	C	S
8	0	3	Mr. Harbeck	male	26	1	0	1	7.12	C	S
9	1	1	Mr. Hopper	male	41	1	0	1	13.00	C	S
10	1	3	Mr. Ingham	male	27	1	0	1	7.25	C	S
- Cell 3:** Displays the last 10 rows of the dataset.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
891	0	3	Mr. O'Connell	male	32	1	0	1	12.00	C	S
892	1	1	Mr. Quinn	male	35	1	0	1	15.50	C	S
893	1	3	Mr. Tremain	male	29	1	0	1	7.25	C	S
894	1	1	Mr. Warkentin	male	30	1	0	1	15.50	C	S
895	1	3	Mr. Williams	male	26	1	0	1	7.25	C	S
896	1	1	Mr. Wood	male	30	1	0	1	15.50	C	S
897	1	3	Mr. Yount	male	26	1	0	1	7.25	C	S
898	1	1	Mr. Zetterstrom	male	30	1	0	1	15.50	C	S
899	1	3	Mr. Brown	male	26	1	0	1	7.25	C	S
900	1	1	Mr. Fisher	male	30	1	0	1	15.50	C	S
- Cell 4:** Displays the first 10 rows of the dataset with the 'Embarked' column converted to numeric values.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund	male	22	1	0	1	5.49	C	0
2	1	1	Mr. Allen	male	35	1	0	24	53.1	C	0
3	1	3	Mr. Borchers	male	39	1	0	33	51.66	C	0
4	1	1	Mr. Brown	male	54	1	0	1	5.00	C	0
5	0	3	Mr. Carter	male	20	1	0	1	5.28	C	0
6	1	1	Mr. Fisher	male	30	1	0	1	12.36	C	0
7	1	3	Mr. Gibson	male	25	1	0	1	7.25	C	0
8	0	3	Mr. Harbeck	male	26	1	0	1	7.12	C	0
9	1	1	Mr. Hopper	male	41	1	0	1	13.00	C	0
10	1	3	Mr. Ingham	male	27	1	0	1	7.25	C	0
- Cell 5:** Displays the last 10 rows of the dataset with the 'Embarked' column converted to numeric values.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
891	0	3	Mr. O'Connell	male	32	1	0	1	12.00	C	0
892	1	1	Mr. Quinn	male	35	1	0	1	15.50	C	0
893	1	3	Mr. Tremain	male	29	1	0	1	7.25	C	0
894	1	1	Mr. Warkentin	male	30	1	0	1	15.50	C	0
895	1	3	Mr. Williams	male	26	1	0	1	7.25	C	0
896	1	1	Mr. Wood	male	30	1	0	1	15.50	C	0
897	1	3	Mr. Yount	male	26	1	0	1	7.25	C	0
898	1	1	Mr. Zetterstrom	male	30	1	0	1	15.50	C	0
899	1	3	Mr. Brown	male	26	1	0	1	7.25	C	0
900	1	1	Mr. Fisher	male	30	1	0	1	15.50	C	0

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.
2. Convert the 'Sex' column to numeric values (male: 0, female: 1).
3. One-hot encode the 'Embarked' column, dropping the first category.
4. Get the mean age of passengers.
5. Get the median fare of passengers.
6. Get the number of passengers by class.
7. Get the number of passengers by gender.
8. Get the number of passengers by survival status.
9. Calculate the survival rate of passengers.
10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
Sample Data:											

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

--	--	--	--	--	--	--	--	--	--	--	--

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

CODE:

```
import pandas as pd
import
```

```
numpy as np
```

```
# Load the Titanic dataset data =
```

```
pd.read_csv('Titanic-Dataset.csv')
```

```
data['FamilySize'] = data['SibSp'] + data['Parch']
```

```
# 1. Create a new column 'IsAlone' (1 if alone, 0 otherwise) data['Alone'] =
```

```
np.where(data['FamilySize']==0,1,0)
```

```
# 2. Convert 'Sex' to numeric (male: 0, female: 1) data['Sex'] =
```

```
data['Sex'].map({'male': 0, 'female': 1})
```

```
# 3. One-hot encode the 'Embarked' column data =  
pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# 4. Get the mean age of passengers mean_age =  
data['Age'].mean() print(mean_age)
```

```
# 5. Get the median fare of passengers median_fare =  
data['Fare'].median() print(median_fare)
```

```
# 6. Get the number of passengers by class passengers_by_class =  
data['Pclass'].value_counts() print(passengers_by_class)
```

```
# 7. Get the number of passengers by gender passengers_by_gender =  
data['Sex'].value_counts().sort_index() print(passengers_by_gender)
```

```
# 8. Get the number of passengers by survival status passengers_by_survival =  
data['Survived'].value_counts().sort_index() print(passengers_by_survival)
```

```
# 9. Calculate the survival rate survival_rate =  
data['Survived'].mean() print(survival_rate)
```

```
# 10. Calculate the survival rate by gender gender_survival =  
data.groupby('Sex')['Survived'].mean() print(gender_survival)
```

Course

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Average time: 0.057 s, 57.00 ms | Maximum time: 0.057 s, 57.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 57 ms Debug

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3...491	3...491
1...216	1...216
2...184	2...184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0...577	0...577
1...314	1...314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0...549	0...549
1...342	1...342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3838383838383838	0.3838383838383838
Sex	Sex
0...0.188908	0...0.188908
1...0.742038	1...0.742038
Name: Survived, dtype: float64	Name: Survived, dtype: float64

< Prev Reset Submit Next >

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

Emb

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

CODE:

```
import pandas as pd
```

```
import numpy as np
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv') data['FamilySize'] = data['SibSp']  
+ data['Parch'] data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1) data  
= pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# 1. Calculate the survival rate by class print(data.groupby('Pclass')['Survived'].mean())
```

```
# 2. Calculate the survival rate by embarked location  
print(data.groupby('Embarked_S')['Survived'].mean())
```

```
# 3. Calculate the survival rate by family size  
print(data.groupby('FamilySize')['Survived'].mean()) # 4.  
Calculate the survival rate by being alone  
print(data.groupby('IsAlone')['Survived'].mean())
```

```
# 5. Get the average fare by class print(data.groupby('Pclass')['Fare'].mean())
```

```
# 6. Get the average age by class print(data.groupby('Pclass')['Age'].mean())
```

```
# 7. Get the average age by survival status print(data.groupby('Survived')['Age'].mean())
```

```
# 8. Get the average fare by survival status print(data.groupby('Survived')['Fare'].mean()) # 9.
```

```
Get the number of survivors by class print(data[data['Survived'] ==  
1]['Pclass'].value_counts())
```

```
# 10. Get the number of non-survivors by class print(data[data['Survived'] ==  
0]['Pclass'].value_counts())
```

The screenshot shows a Jupyter Notebook interface. On the left, a list of columns from the Titanic dataset is displayed: Survived, Pclass, Name, Sex, Age, SibSp, Parch, Fare, Cabin, Embarked. On the right, a preview of the dataset rows is shown, with columns labeled Survived, Pclass, Name, Sex, Age, SibSp, Parch, Fare, Cabin, Embarked. The data rows show individual passengers with their respective attributes.

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below,

Sample Data:

Emb

PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S

10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C

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Explorer

Average time 0.059 s 59.00 ms Maximum time 0.059 s 59.00 ms 1 out of 1 shown test case(s) passed

Test case 1 59 ms Debug

Expected output	Actual output
female --- 233	female --- 233
male --- 109	male --- 109
Name: Sex, dtype: int64	Name: Sex, dtype: int64
male --- 468	male --- 468
female --- 81	female --- 81
Name: Sex, dtype: int64	Name: Sex, dtype: int64
1 --- 217	1 --- 217
0 --- 125	0 --- 125
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
1 --- 427	1 --- 427
0 --- 122	0 --- 122
Name: Embarked_S, dtype: int64	Name: Embarked_S, dtype: int64
0.5398230088495575	0.5398230088495575
0.3810316139767055	0.3810316139767055
28.0	28.0
28.0	28.0
26.0	26.0
10.5	10.5

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